

Sean Balbale

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EDUCATION

Trinity College — Hartford, CT

Expected May 2027

Bachelor of Science in Electrical and Computer Engineering

GPA: 3.65

Bachelor of Science in Computer Science

Coursework: High-Performance Computing, Embedded Systems Design, Deep Learning, Software Engineering, Microelectronic Circuits, Data Structures & Algorithms, Automatic Control Systems, Intro to Computer Systems.

Extracurriculars: Varsity Athlete — Trinity College Men's Rowing; Singer & Technology Chair — The Trinity College Accidentals.

SKILLS

Languages: Java, Python, C, C++, TypeScript, JavaScript, Assembly (8051/MIPS), Verilog, SQL, MATLAB, LaTeX

Backend: Spring Boot 3, Jakarta EE, JUnit 5, Mockito, REST APIs, Node.js

Frontend: React, Next.js, Tailwind CSS, AngularJS

Cloud & DevOps: AWS, Kubernetes, Docker, Git, Linux, CI/CD, Log4j2, Slurm, MPI

Embedded / Hardware: Arduino, ARM, 8051, FPGA, BLE 5, ANT+, SPI/I2C, LTspice, Oscilloscopes, Logic Analyzers

Other: SolidWorks, 3D Printing (Klipper), Precision Metrology, Systems Engineering

WORK EXPERIENCE

Software Engineer Intern — Sonos | Boston, MA

Summer 2026

- Led migration of two production Java/Spring Boot services (Cloud Content team) from WebSocket command delivery to an SQS-backed fire-and-forget queue: authored the design plan and Jira epic, implemented feature-flagged per-player fan-out with retry-until-acceptance semantics, and validated both transports against real speakers.
- Diagnosed a silent delivery failure in a pre-production environment by tracing commands through the SDK and queue infrastructure to an endpoint-routing defect, and drove the fix across both services.
- Made the service's functional-test suite runnable locally for the first time by porting its DynamoDB-stream Lambda into a local test harness, enabling 24-minute pre-push validation ahead of the hour-long shared CI run; also built Datadog dashboards and monitors for the new delivery path.

Engineering Teaching Assistant — ENGR 212L Linear Circuit Theory, Trinity College | Hartford, CT

Spring 2026

- Led weekly lab sessions for ENGR 212L covering RLC circuit analysis, op-amps, Laplace transforms, Fourier series, and active filter design.

Software Engineering Intern — Bullhorn | Boston, MA

Summer 2025

- Owned BH Boot 1.0.0 framework migration: upgraded multiple production Java services from Spring Boot 2 to Spring Boot 3, refactoring all `javax` namespaces to `jakarta` and restructuring the proprietary data access layer.
- Led Log4j2 security upgrade across `SEARCH_AI`, `ATS_CRM`, and `MIDDLE_OFFICE` groups; refactored legacy TestNG/JMockit test suites to JUnit 5 and Mockito; managed feature branches, merge requests, and peer code reviews.

Metrology Intern — East Coast Metrology, A Division of In-Place Machining Co. | Topsfield, MA

Summer 2024

- Calibrated portable CMMs, total stations, laser trackers, and large-volume scanners; verified manufactured parts against SolidWorks CAD tolerances in-house and on-site.

RESEARCH, HONORS & SELECTED PROJECTS — MORE ON PORTFOLIO WEBSITE

"Embodied Harms and Inferred Data: Redefining Privacy in Extended Reality"

Expected Late August 2026

The International Journal of Law, Ethics, and Technology — In Production

- Peer-reviewed paper arguing BIPA and GDPR are structurally inadequate for XR environments; proposes a "Bodyright" framework reclassifying biometric data as an inalienable extension of human sovereignty. Awarded the **Alumni Prize in English Composition**, Trinity College, May 2026.

Edge-Compute Pacing Node — Senior Capstone (CPSC 498 / ENGR 483)

Fall 2026 – Spring 2027

- Embedded node on Nordic nRF52840 running W' balance fatigue model in C/C++ on FreeRTOS; ≤ 100 ms sensor-to-LED latency via BLE 5/ANT+ — entirely offline, no cloud or smartphone offload.

Personal Portfolio — seanbalbale.com

Spring 2026

- Production Next.js/React/TypeScript app on Vercel with CI/CD; custom backend API for transactional email dispatch, dark/light mode theming, and embedded PDF viewer.

PERSONAL

Serious home cook across a wide range of cuisines — a proper katsu ramen takes hours and a good samosa crimp takes thirty seconds of muscle memory. Outside the kitchen: cycling, ski mountaineering, and tinkering on whatever project has captured my attention that week.